```
!pip uninstall -y transformers
!pip install -U transformers accelerate bitsandbytes sentencepiece
```

```
Found existing installation: transformers 5.12.0
Uninstalling transformers-5.12.0:
  Successfully uninstalled transformers-5.12.0
Collecting transformers
  Using cached transformers-5.12.0-py3-none-any.whl.metadata (33 kB)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.14.0)
Requirement already satisfied: bitsandbytes in /usr/local/lib/python3.12/dist-packages (0.49.2)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.25.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.60 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.11.0+cu128)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (7
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,c
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->ac
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
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Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->acc
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->a
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-t
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
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Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
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Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-tool
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[c
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer-
```

```
import torch
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig
import pandas as pd
import re
```

```
def extract_final_num(text):
    match = re.search(r"Final\s+Answer:\s*(-?\d+\.?\d*)", text, re.IGNORECASE)
    if match:
        return match.group(1).strip('.')
    nums = re.findall(r'-?\d+\.?\d*', text)
    if nums:
        return nums[-1].strip('.')
    return None
```

```
test_df = pd.read_csv("unified_svamp_test.csv").head(50)
```

```
import os
os.environ['LD_LIBRARY_PATH'] += ':/usr/local/cuda-13.0/lib64'
```

```
from google.colab import userdata
from huggingface_hub import login
```

```
hf_token = userdata.get('hf_token')
login(hf_token)
print("Authenticated successfully!")
```

```
Authenticated successfully!
```

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig
model_id = "microsoft/Phi-3-mini-4k-instruct"
bnb_config = BitsAndBytesConfig(
    load_in_4bit = True,
    bnb_4bit_quant_type = "nf4",
    bnb_4bit_compute_dtype=torch.float16
)
```

```
tokenizer = AutoTokenizer.from_pretrained(model_id, add_bos_token=False)
stop_tokens = ["<|end|>", "<|endoftext|>"]
eos_token_ids = [tokenizer.convert_tokens_to_ids(t) for t in stop_tokens if tokenizer.convert_tokens_to_ids(t) is not None]
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

```
print("Downloading the model")
model = AutoModelForCausalLM.from_pretrained(model_id, quantization_config=bnb_config, device_map="auto")
print("Sucess! The engine is running")
```

```
Downloading the model
Loading weights: 100%                                  195/195 [00:30<00:00,  7.17it/s]
Sucess! The engine is running
```

```
results = []
correct_count = 0
total_count = len(test_df)
```

```
print("\nStarting Baseline Evaluation-->")
for index, row in test_df.iterrows():
    question = row['question']
    ground_truth = str(row['answer']).strip()
    messages = [
    {"role": "user", "content": f"You are an expert math tutor. Question: {question}\nAnswer the question step-by-step. You
    ]
    prompt = tokenizer.apply_chat_template(messages, tokenize=False, add_generation_prompt=True)
    inputs = tokenizer(prompt, return_tensors="pt").to("cuda")
    outputs = model.generate(
        **inputs,
        max_new_tokens=300,
        temperature=0.1,
        do_sample=False,
        eos_token_id=eos_token_ids,
    )
    response = tokenizer.decode(outputs[0][inputs['input_ids'].shape[-1]:], skip_special_tokens=True)
    extracted_answer = extract_final_num(response)

    is_correct = False
    if extracted_answer is not None:
        try:
            if float(extracted_answer) == float(ground_truth):
                is_correct = True
                correct_count += 1
        except ValueError:
            pass

    print(f"\n--- Problem {index + 1} ---")
    print(f"Question: {question}")
    print(f"Model Output:\n{response.strip()}")
    print(f"Extracted Answer: {extracted_answer}")
    print(f"Ground Truth Answer: {ground_truth}")
    print(f"Correct: {is_correct}")
    print("-" * 30)
```

```
Step 3: Determine the depth of the water.
The depth of the water is 2 times Dean's height, so we multiply Dean's height by 2:
6 feet * 2 = 12 feet.

Final Answer: 12.
Extracted Answer: 12
Ground Truth Answer: 12
Correct: True
------------------------------

--- Problem 49 ---
Question: Every day Ryan spends 6 hours on learning english and 7 hours on learning chinese. If he learns for 5 days How m
Model Output:
Step 1: Calculate the total hours spent on learning English per day.
Ryan spends 6 hours on learning English every day.

Step 2: Calculate the total hours spent on learning Chinese per day.
Ryan spends 7 hours on learning Chinese every day.

Step 3: Calculate the total hours spent on learning both English and Chinese per day.
Total hours spent per day = Hours spent on English + Hours spent on Chinese
Total hours spent per day = 6 hours (English) + 7 hours (Chinese)
Total hours spent per day = 13 hours

Step 4: Calculate the total hours spent on learning both English and Chinese over 5 days.
Total hours spent over 5 days = Total hours spent per day * Number of days
Total hours spent over 5 days = 13 hours * 5 days
Total hours spent over 5 days = 65 hours

Final Answer: 65
Extracted Answer: 65
Ground Truth Answer: 65
Correct: True
------------------------------

--- Problem 50 ---
Question: Jack received 10 emails in the morning, 5 emails in the afternoon and 4 emails in the evening. How many more ema
Model Output:
Step 1: Identify the number of emails received in the afternoon.
Jack received 5 emails in the afternoon.

Step 2: Identify the number of emails received in the evening.
Jack received 4 emails in the evening.

Step 3: Calculate the difference between the number of emails received in the afternoon and the evening.
Difference = Emails in the afternoon - Emails in the evening
Difference = 5 - 4

Step 4: Calculate the final answer.
Difference = 1

Final Answer: 1.
Extracted Answer: 1
Ground Truth Answer: 1
Correct: True
------------------------------
```

```python
accuracy = (correct_count / total_count) * 100
print(f"FINAL BASELINE ACCURACY: {accuracy:.2f}% ({correct_count}/{total_count})")
```

```
FINAL BASELINE ACCURACY: 82.00% (41/50)
```